

Beyond Persistence? PAGE: Persistence-Anchored Gated Forecasting with Spatio-Temporal Graph Networks for Dengue

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Abstract—[WRITE: abstract, 180–220 words, written last]

Index Terms—dengue forecasting, spatio-temporal graph neural networks, persistence, SEIR, data leakage, prospective evaluation

I. INTRODUCTION

[WRITE: P1 background]

[WRITE: P2 existing work]

[WRITE: P3 the technical gap]

[WRITE: P4 PAGE and the main results]

[WRITE: P5 three contributions, numbered]

II. RELATED WORK

[WRITE: (a) ST-GNNs and the gap]

[WRITE: (b) mechanistic hybrids and the gap]

[WRITE: (c) persistence and simple baselines, link to PAGE]

III. DATASET

[WRITE: (a) what is wrong with the benchmark array]

[WRITE: (b) the corrected 559-week dataset]

[WRITE: (c) the 2024–2026 extension, with the amendment stated]

IV. THE PAGE FRAMEWORK

[WRITE: (a) problem formulation, symbols only]

[WRITE: (b) encoder]

[WRITE: (c) the PAGE head: equation and why it works]

[WRITE: (d) the optional SEIR branch, 2–3 sentences]

[WRITE: (e) loss]

[WRITE: (f) implementation details, with the numbers]

V. EXPERIMENTAL SETUP

[WRITE: (a) baselines]

[WRITE: (b) protocol, stated as a fair comparison]

[WRITE: (c) metrics and statistical tests]

VI. RESULTS

A. Comparative analysis

[WRITE: (a) Table encoders: who fails, how PAGE repairs, against persistence]

B. Ablation study

[WRITE: (b) Table rescue: anchor, gate, SEIR branch; why the pure SEIR decoder fails]

TABLE I

COMPARATIVE ANALYSIS ON ONE HARNESS (CORRECTED DATA, 3 ORIGINS $\times$ 3 SEEDS). MEAN VALIDATION / TEST RMSE IN WEEKLY CASES. TOP: THE FIVE PUBLISHED ST-GNNs AND THE LSTM OF SEIR-LSTM, AS PUBLISHED (DIRECT HEAD), WITH PAGE, AND WITH PAGE PLUS THE SEIR BRANCH (THE SEIR-GNN, OR SEIR-LSTM FOR THE LSTM). SQUARED ERROR, NO SEASONAL FEATURES. BOTTOM: BASELINES; B IS THE CONFIGURATION VALIDATION SELECTS. SELECTION USES VALIDATION ONLY.

encoder	direct	PAGE	PAGE+SEIR
AAGCN	16.71 / 35.15	16.57 / 33.53	17.43 / 35.58
ASTGCN	16.83 / 35.61	16.76 / 34.76	17.51 / 35.71
LSTM	16.91 / 35.10	16.94 / 35.85	17.69 / 35.52
STGAT	22.93 / 62.00	16.98 / 36.46	17.63 / 35.84
A3TGCN	28.63 / 60.40	17.68 / 37.16	17.68 / 35.50
DCRNN	34.64 / 73.52	18.35 / 39.02	17.67 / 35.52
baseline		validation / test	
Persistence		17.86 / 36.02	
No message passing (res.)		16.90 / 34.57	
GCN (res.)		16.84 / 34.50	
STID-style MLP [1]		16.35 / 36.33	
k -NN analogues		16.52 / 34.34	
NB-GLM		16.97 / 42.85	
Boosted trees		17.94 / 38.96	
Boosted trees + climate		17.40 / 36.49	
B : AAGCN, NB, season		15.51 / 37.54	

C. Hyperparameter tuning and computational cost

[WRITE: (c) tuning and (d) cost, a few sentences]

D. Prospective evaluation, 2024–2026

[WRITE: (e) Table prospective, stated honestly]

VII. DISCUSSION

[WRITE: why PAGE works, its limit, physics, limitations]

VIII. CONCLUSION

[WRITE: 3–4 sentences and the code link]

REFERENCES

- [1] Z. Shao, Z. Zhang, F. Wang, W. Wei, and Y. Xu, “Spatial-temporal identity: A simple yet effective baseline for multivariate time series forecasting,” in *ACM International Conference on Information and Knowledge Management (CIKM)*, 2022.

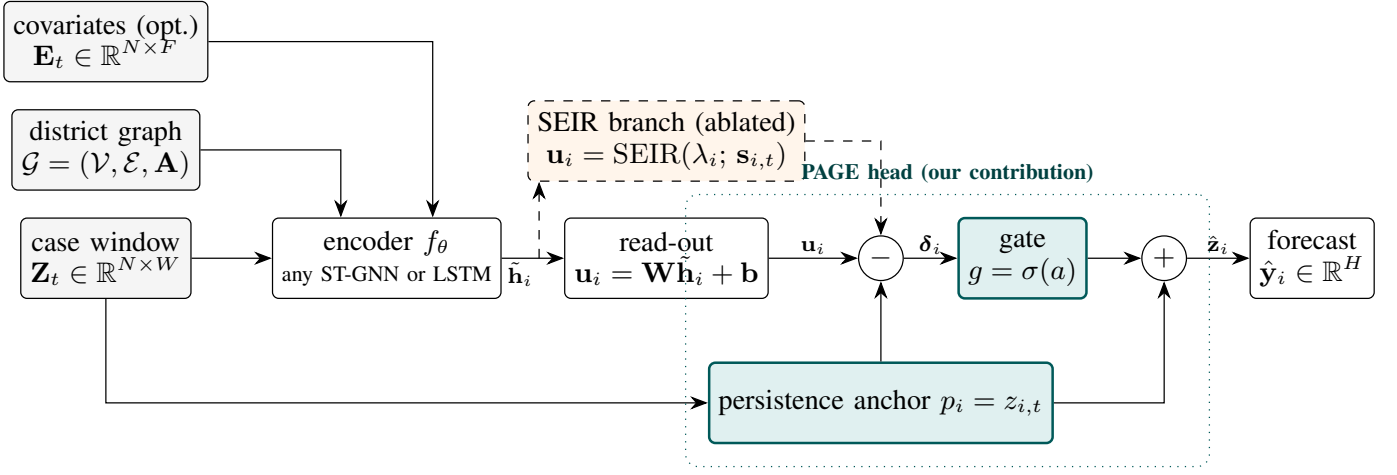


Fig. 1. PAGE. An unchanged encoder reads the last W weeks of standardised log cases on the district graph. Its read-out $\mathbf{u}_i$ is turned into a departure $\delta_i = \mathbf{u}_i - p_i \mathbf{1}$ from the persistence anchor, shrunk by a learned gate, and added back to the anchor. The dashed SEIR branch replaces the read-out with a differentiable simulator; it is the component the ablation removes.

TABLE II

ABLATION: FROM THE PUBLISHED DIRECT HEAD TO PAGE. EACH ENCODER BEHIND FOUR HEADS THAT ADD, IN TURN, THE PERSISTENCE ANCHOR (RESIDUAL), THE LEARNED GATE (PAGE) AND THE SEIR BRANCH (MEAN VALIDATION RMSE; CONFIGURATION OF TABLE I). THE LAST TWO COLUMNS PAIR PAGE+SEIR WITH PAGE; NEGATIVE FAVOURS THE SEIR BRANCH, AND THE COUNT IS RUNS WHERE IT WINS.

encoder	direct	resid.	PAGE	SEIR branch		
				PAGE	+SEIR	
					val	test
AAGCN	16.71	16.72	16.57	17.43	+0.86 (1/9)	+2.05 (0/9)
ASTGCN	16.83	16.85	16.76	17.51	+0.75 (0/9)	+0.95 (3/9)
LSTM	16.91	16.96	16.94	17.69	+0.75 (0/9)	-0.33 (6/9)
STGAT	22.93	17.85	16.98	17.63	+0.66 (1/9)	-0.62 (8/9)
A3TGCN	28.63	17.47	17.68	17.68	-0.00 (6/9)	-1.66 (9/9)
DCRNN	34.64	17.52	18.35	17.67	-0.68 (3/9)	-3.50 (9/9)
persistence		17.86				

TABLE III

PROSPECTIVE EVALUATION ON WEEKS NO MODEL OR SETTING HAD SEEN: 85 FORECAST STARTS, 2024-W11 TO 2026-W32. SETTINGS FROZEN ON DEVELOPMENT DATA BEFORE THE NEW WEEKS WERE PARSED. NEURAL ARMS: MEAN OVER 3 SEEDS, SEED RANGE OF RMSE IN BRACKETS. **BOLD: BEST PER METRIC.** PRE-REGISTERED TESTS (BLOCK BOOTSTRAP OVER FORECAST STARTS, HOLM OVER TWO): P1, SEIR BRANCH VS PAGE: $-0.03 [-0.41, +0.12]$, $p_{\text{HOLM}} = 0.67$; P2, PAGE+SEIR VS PERSISTENCE: $+0.65 [-0.99, +1.69]$, $p_{\text{HOLM}} = 0.62$.

arm	RMSE	MAE	SMAPE	MAPE	RMSE $h=1/2/3$
Persistence	34.75	11.78	50.3	53.3	15.4/35.7/46.0
PAGE (adaptive graph)	35.43 [35.24, 35.60]	11.35	45.7	43.9	16.0/35.7/47.3
PAGE + SEIR branch	35.40 [35.11, 35.56]	11.29	46.1	43.5	15.8/35.7/47.3
Residual head (no gate)	35.74 [35.37, 36.19]	11.61	46.6	43.0	16.4/36.0/47.6
AR(3) ridge	36.35	11.35	46.7	44.5	17.3/35.9/48.8
Seasonal naive	62.43	27.53	74.6	113.4	50.7/62.5/72.2